

Interview with Riya Rasika and Kishor Kumar

Hi Riya. Congrats for winning Aurora Energy Award for her project "*The Strength Within: Te ara hauora Māori*".

Qn: Can you introduce yourself?

Kia Ora, my name is Riya Rasika Kishor Kumar. I am a Year 10 student at Columba College in Dunedin, New Zealand. I enjoy learning about science and how it can be used to solve real-world problems. I am particularly interested in health and research because I find it fascinating how evidence and investigation can help improve people's lives.

Qn: Can you tell us about your hobbies/interests?

My hobbies include science research, badminton, dance, swimming, art and learning new things. I enjoy exploring topics related to health and biology. I also like setting personal goals and challenging myself to improve both academically and personally.

Qn: What motivates you to participate in science experiments ?

I enjoy science experiments because they allow me to investigate questions and discover answers through evidence. I like the process of testing ideas, analysing results, and learning from mistakes. It is exciting to know that even a small study can contribute to a better understanding of a problem and potentially help others.

Qn: Can you share the experience of collaborating with your dad ?

Working with my dad was one of the most valuable parts of this project. He supported me throughout the research process and helped me think more critically about the data we collected. One area where he particularly helped was coding in Python. He used Python to assist with data analysis and visualisation, and through this process I learned how coding can be used as a powerful tool in scientific research.

Watching and discussing the code with him helped me better understand how data can be organised, analysed, and presented clearly. It also showed me how technology and science can work together to answer research questions. Beyond the technical aspects, my dad encouraged me to stay curious, ask questions, and keep improving the project. Working together taught me the importance of teamwork, problem-solving, and continuous learning.

Qn: At a high level can you give us an overview on how to conduct a study?

1. Identify a Question or Problem

Start by choosing a topic and developing a clear research question that you want to investigate.

2. Conduct Background Research

Read books, articles, and reliable sources to understand what is already known about the topic.

3. Plan the Method

Design a fair and reliable way to collect information or data that will help answer the research question.

4. Collect Data

Carry out the study by gathering observations, measurements, survey responses, or other relevant data.

5. Analyse the Data

Organise and examine the data to identify patterns, trends, or relationships.

6. Interpret the Results

Consider what the findings mean and whether they answer the original research question.

7. Draw Conclusions

Summarise the key findings and explain what was learned from the study.

8. Communicate the Findings

Share the results through a report, presentation, poster, or publication so that others can learn from the research.

Throughout all these steps, it is important to remain objective, use evidence to support conclusions, and be open to learning from both expected and unexpected results.

Qn: What are some of the pain points of doing a scientific study?

One challenge is collecting reliable data, especially when resources or time are limited. Another difficulty is dealing with unexpected results, which may require revising the method or repeating parts of the study. Research also requires patience because analysing information and drawing accurate conclusions can take a significant amount of time and effort.

Qn: If you would ask for support from the School or Government for research, what would that be?

I would appreciate greater access to research opportunities, mentorship programmes, and scientific resources for students. It would be helpful if schools could provide more opportunities for students to participate in research projects and connect with scientists and university researchers. Support for science fairs, competitions, and community-based research initiatives would also encourage more young people to explore science and innovation.